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## **An Interactive Python Based Tool for Flow Rate Prediction and Express Well Completion Design**

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### **Abstract**

Accurately predicting oil flow rate in the presence of near-wellbore damage remains a persistent challenge in well completion design, especially when multiple subsurface and fluid properties must be considered. This study introduces SkinFlow, an interactive Python-based application that helps evaluate flow efficiency in open-hole completions for both vertical and horizontal wells. Using established analytical models, the platform lets users adjust input parameters and instantly see the effect on skin factor and oil production rate.

SkinFlow offers real-time sensitivity analysis, combined-parameter evaluation, and Monte Carlo simulation. Engineers can test the effect of individual variables or apply several changes together, mimicking treatments such as acidizing, damage removal, or wettability alteration. Probabilistic modeling allows uncertainty in key parameters to be represented through probability distributions, giving a realistic range of outcomes.

To demonstrate its value, a case study compared two wells in the same reservoir. Well №2026 produced 500.1 STB/D with a skin of 7.19, while Well №2025, affected by severe damage, produced only 268.3 STB/D with a skin of 20.72. Deterministic stimulation modeling of Well №2025, with altered-zone permeability raised from 10 to 200 mD, increased production to 1207 STB/D and reduced the skin by 43.5 percent. Monte Carlo analysis confirmed clear gains, forecasting post-stimulation rates with a median of 981 STB/D.

A second application focused on the pilot horizontal Well №872\_H, drilled into a tight, low-permeability formation. Using SkinFlow, the stimulation team was able to rapidly explore how changes in drawdown and near-wellbore permeability would influence performance. The platform delivered an express forecast of potential gains, showing that even under conservative assumptions the well could increase from about 120 STB/D to around 840 STB/D, giving the team confidence to move ahead with stimulation.

These results show that SkinFlow can guide stimulation planning while also serving as a practical educational tool. Students can interact with real reservoir parameters and open-hole completion characteristics, adjust values in real time, and instantly see how those changes impact oil rate and formation

damage. This hands-on interaction helps students build a clear understanding of how well performance is shaped by completion design, reservoir conditions, and stimulation.

**Keywords:** Well Completion Design, Flow Rate Prediction, Skin Factor Analysis, Python, Formation Damage

## Introduction

The last stage in well construction is known as well completion. It is an important link in oil field development between the productive formation and the wellbore; the design level and implementation quality play a key role in the expected production target and economic effectiveness of any oil field (Furui, Zhu and Hill, 2004; Velayati *et al.*, 2018; Abobaker *et al.*, 2022). Selection of completion started with determining several parameters such as earth stresses, rock mechanical properties, reservoir pressure, temperature, sand production, permeability, formation damage, reservoir thickness, formation type such as sandstone, and formation stability over the producing life of the well (Furui *et al.*, 2007). Completion design should not only meet the requirements of flowing production of an oil well, but consider the requirements of artificial lift production in later stage, and also create favorable condition of necessary downhole operations and measurement, an appropriate well completion design would lead to reduce well cost, enhanced hydrocarbon production, minimized water breakthrough, but more importantly, improved well performance and ultimate hydrocarbons recovery (Civan, 2007; Ashena *et al.*, 2021). There are several different types of well completion, and fluid flow can be significantly affected by the type of well completion, including, open hole, slotted linear, cased and perforated, sand control (sand-screens, pre-packed screens, gravel packing) (Joseph and Ike, 2024). The selection of completion methods becomes especially critical in horizontal wells due to their extensive reservoir contact, where fluid enters the wellbore at varying locations and flow rates along the horizontal section (Velayati *et al.*, 2018). Similarly, horizontal well damage can have a severe impact on the well production and can cause a considerable reduction from the ideal productivity index. Also, horizontal wells are easy to encounter premature water and gas breakthrough (Retnanto and Yamin, 1999). If well completion is not executed properly or the incorrect completion is selected, there will be a drastic drop in pressure, and this added pressure across the wellbore completion is known as completion skin (Velayati *et al.*, 2018; Abobaker *et al.*, 2022). The decline and impairment of natural inherent productivity and injectivity of oil and gas wells at any point in time due to drilling, completion, production, stimulation, kill or workover activities is called formation damage (Mahmoudi *et al.*, 2017; Yuan and Wood, 2018). Skin may be due to mechanical or turbulent effects. The higher the skin, the larger the pressure drawdown and the lower the well production would be. Mechanical skin is mainly because of completion and independent of flow rate (Ouyang and Huang, 2005). Porous medium involves many rate processes that occur at rates of different orders of magnitude. Therefore, understanding the impact of the rate on in situ rock and fluid and the interaction between fluid and rock is important to control formation damage (Bennion, 2002; Civan, 2007). Understanding formation damage is essential for accurate diagnosis, effective prevention, and successful well stimulation, as it encompasses key aspects such as damage prevention, assessment, control, and removal (Krueger, 1986).

Although the software developed in this study is capable of computing formation damage effects for both vertical and horizontal wells, the following sections will specifically provide a brief discussion on formation damage and its impact on horizontal well productivity.

## Formation Damage and Its Impact on Horizontal Well Productivity

Any stage of a well's lifecycle, including drilling, completion, production, and workover operations, can result in formation damage. Understanding the mechanisms, forms, and causes of formation degradation has grown more crucial as exploitation activities become more intricate and concentrate on deeper and tighter

reservoirs. This is because well productivity is severely harmed by formation damage, especially in the vicinity of the wellbore, which is crucial for open-hole completions (Patel and Singh, 2016). Due to the maximize reservoir exposure and increasing well production, horizontal wells have become increasingly popular worldwide. These wells allow for the exploitation of thin pay zones, the targeting of several zones, and the reduction of drawdowns to reduce early water or gas coning. Recently, horizontal wells have also been employed as producers and injectors in secondary and tertiary enhanced oil recovery (EOR) processes and steam-assisted gravity drainage (Ezenweichu and Laditan, 2015).

Horizontal wells have been drilled more often all over the world in recent years. A horizontal well's primary goal is to increase reservoir contact, which in turn raises well productivity, a crucial component of EOR applications. While a vertical well crosses the bedding plane at a 90° angle, a horizontal well is drilled parallel to the reservoir's bedding plane. To maximize reservoir contact and enhance flow performance, horizontal wells have a greater exposure to surface area than vertical wells (Konwar *et al.*, 2023). Notwithstanding its benefits, a lot of horizontal wells have had unsatisfactory results, and inferior flow performance is frequently mostly caused by formation damage in the vicinity of the wellbore. This is particularly true when the damage near the wellbore prevents fluid flow even when the reservoir quality is viable. Even relatively shallow, invasive near-wellbore damage, which would normally be avoided in cased and cemented vertical completions, can seriously impair fluid flow because the majority of horizontal wells are completed in an open-hole technique (Paköz *et al.*, 2024; SUN *et al.*, 2024).

In general, high-permeability reservoirs exhibit less formation damage than low-permeability ones. Because horizontal wells expose a bigger reservoir than vertical wells, the skin factor, which measures formation damage, frequently has less of an effect on well productivity in horizontal wells. However, compared to vertical wells, horizontal wells may sustain more near-wellbore damage due to the longer drilling times involved. To reduce or clean up the damage, appropriate processes must be put in place (Li *et al.*, 2023; Zhang *et al.*, 2023; Moen, Antonsen and Ejofodomi, 2024; Xu *et al.*, 2024).

The skin factor ( $S$ ), a dimensionless parameter representing additional pressure loss (referred to as  $P_{\text{skin}}$ ), is frequently used to quantify the extent of damage in the bottom-hole zone. The presence of formation damage is indicated by a positive skin factor, and the degree of this damage is correlated with the skin factor's magnitude. Conversely, a negative skin factor indicates improved flow and reservoir contact by demonstrating an increase in the well's effective radius following stimulation (Sultanova, 2022; Al-Rbeawi, 2024; Thabet *et al.*, 2024). As a result, even while horizontal wells have a lot to offer in terms of productivity and reservoir exposure, they are vulnerable to damage to the near-wellbore formation, which can drastically lower well performance. It is significant to highlight that understanding the mechanisms that underlies the damage, causes, and the techniques to diminish it will require maintaining the effectiveness of the horizontal wells, specifically in low-permeability reservoirs (Huang *et al.*, 2024; Yan *et al.*, 2024).

Skin is defined as a dimensionless factor that is calculated to compare the actual state of the pressure drop near the wellbore to an ideal state to evaluate the production efficiency. Joseph and Ike (2022) stated that any deviation of the flow and pressure from the ideal radial flow to the wellbore can be mathematically explained through the skin concept. Fig. 1 shows the pressure profiles in the vicinity of the wellbore for both a well with formation damage and an ideal well while the horizontal well scheme is shown in Fig. 2.

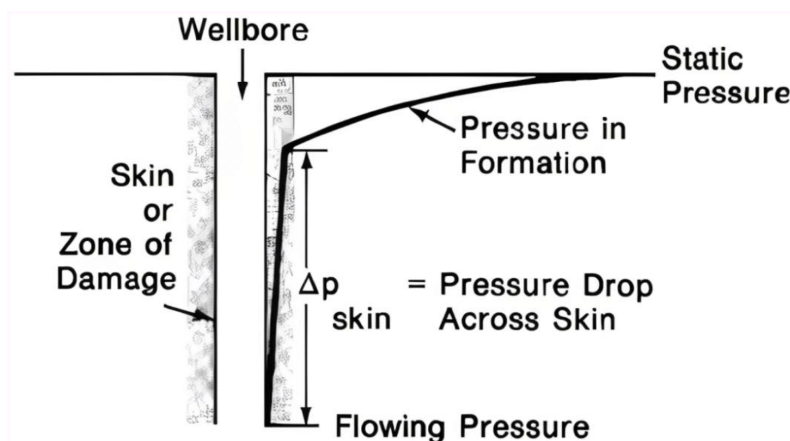


Figure 1—Pressure profile in the near-wellbore region for an ideal well and a well with formation damage (Ezenweichu and Laditan, 2015).

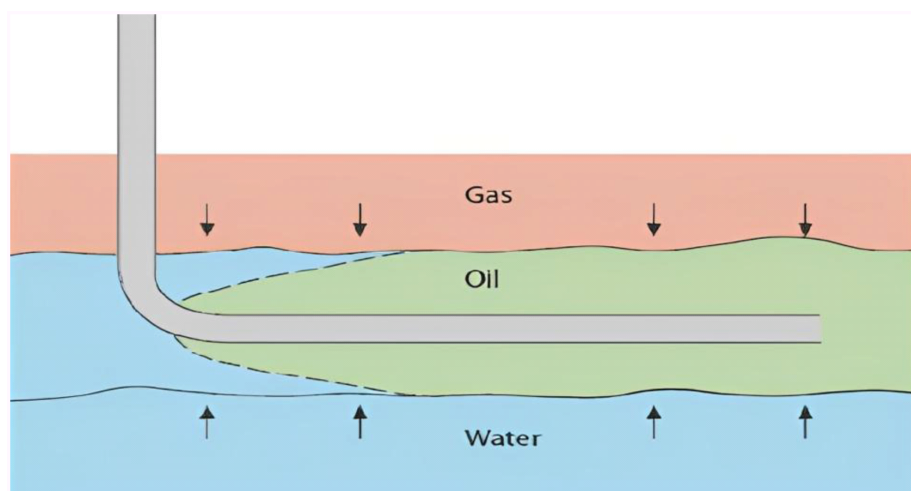


Figure 2—Schematic of a horizontal oil well intersecting a reservoir with a typical gas–oil–water contact (Okon, Appah and Akpabio, 2017).

Skin factor models are available for a variety of horizontal well completions, such as cased and perforated well completions, perforated liner completions, and slotted liner completions (Cooper and Troncoso, 1988; Tang *et al.*, 2000; Furui, Zhu and Hill, 2005; Furui *et al.*, 2007; Dastgerdi, Manshad and Mohammadi, 2020; Roostaei *et al.*, 2021; Joseph and Ike, 2022). While skin factor models exist for various horizontal well completions, each completion type has its own characteristics and implications for well performance. Recognizing these differences is important, and exploring each in depth would be beyond the scope of this paper. Therefore, this study will specifically focus on open hole completions, discussing their unique aspects and how they influence overall productivity.

### A Skin Factor Model for Open Hole

One important metric for open hole completions is the skin factor, which quantifies the additional pressure drop caused by fluid flow restrictions around the wellbore (Yildiz, 2008). This could be caused by a variety of factors, such as mud invasion, formation damage, and inadequate cleaning. The skin factor is typically simulated using the permeability difference between the undamaged reservoir and the damaged near-wellbore zone. Additionally, it considers the effects of convergent flow into the wellbore, which directs flow from a larger area into the well, resulting in increased pressure losses.

To predict the performance of open hole completions, a number of comprehensive skin factor models have been developed. The effects of formation damage and the resulting turbulent flow are all considered



in these models. For instance, there was developed a model that considers the connection between flow dynamics in slotted liners and formation damage, showing how these connections significantly impact well productivity. Furthermore, the model considers the influence of the pressure drop across perforations and turbulent flow, both of which significantly contribute to the apparent skin factor in these completions (An and Wu, 2014; Joseph and Ike, 2024).

The model provides a tool for open hole completion design optimization by accounting for these variables, ensuring that skin effects are minimized to enhance well performance (Furui, Zhu and Hill, 2005). Therefore, to improve reservoir management and well design, it is essential to understand and simulate the skin component in open-hole completions. Accurately forecasting skin effects helps to maximize fluid flow, reduce formation damage, and ultimately boost the economy. Eq. 1 represents the skin model for an open hole vertical well (Hawkins, 1956):

$$S = \left( \frac{k}{k_s} - 1 \right) \ln \left( \frac{r_s}{r_w} \right) \quad (1)$$

Where the  $S$  denotes the skin factor for open hole,  $k$  is the reservoir permeability (mD),  $k_s$  is the permeability of the altered zone (mD),  $r_s$  represents the radius of the altered zone (m or ft), and  $r_w$  represents the wellbore radius (m or ft).

Considering the complexity of factors influencing the skin factor in open hole completions, practical tools that allow engineers to quickly evaluate and optimize these parameters become highly valuable. To address this need, the following chapter introduces an interactive computational platform designed specifically to simulate and assess skin factor impacts on well performance. The methodology highlights how this platform enables engineers and decision-makers to efficiently explore multiple scenarios, refine their completion designs, and ultimately achieve enhanced reservoir management and productivity.

## Methodology

The decision-support tool implemented in this study was developed using Python to assist petroleum engineers in assessing near-wellbore conditions and their impact on skin factor and oil flow rate for the open hole completion (Fig. 3).

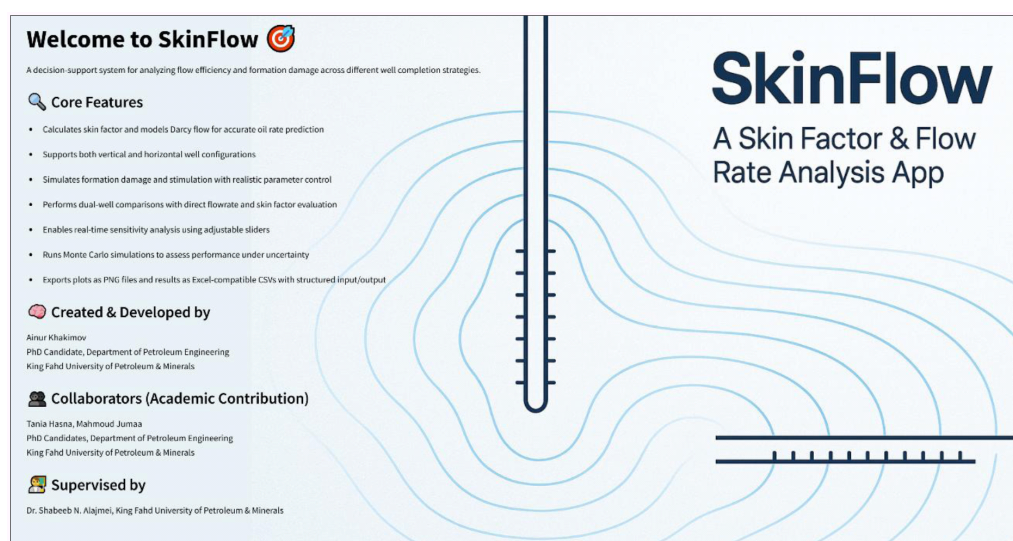


Figure 3—Home page of SkinFlow with core features and contributors.

Built as an interactive interface with real-time feedback, the system supports both vertical and horizontal wells. Users enter reservoir, fluid, and completion parameters through a structured input layout, with all

values stored in memory via a session state structure. This design ensures seamless parameter passing between modules during evaluation and analysis (Fig. 4).

Enable Two-Well Comparison?  
☐ No ☒ Yes

**Well Name №1**  
 Enter Name for Well 1  
 Well №2025 (Block №8)

**Well Name №2**  
 Enter Name for Well 2  
 Well №2026 (Block №9)

Select Well Type  
☒ Vertical ☐ Horizontal

Select Well Type  
☒ Vertical ☐ Horizontal

**Well Completion Type**  
 Select Well Completion Type  
 Open Hole

**Well Completion Type**  
 Select Well Completion Type  
 Open Hole

**Reservoir & Fluid Parameters**

| Parameter                                  | Value   | Adjust |
|--------------------------------------------|---------|--------|
| Reservoir Permeability [k] (mD)            | 75.00   | - +    |
| Permeability of the altered zone [ks] (mD) | 10.00   | - +    |
| Wellbore Radius [rw] (ft)                  | 0.33    | - +    |
| Radius of the altered zone [rs] (ft)       | 8.00    | - +    |
| Reservoir Thickness [h] (ft)               | 25.00   | - +    |
| Reservoir Pressure [p_res] (psi)           | 3500.00 | - +    |
| Well Flowing Pressure [p_wf] (psi)         | 1000.00 | - +    |
| Oil Viscosity [μo] (cP)                    | 3.00    | - +    |
| Oil Relative Permeability (fraction)       | 0.85    | - +    |
| Formation Volume Factor [Bo] (res bbl/STB) | 1.20    | - +    |
| Drainage Radius [re] (ft)                  | 1600.00 | - +    |

Figure 4—Dual-well input interface showing parameters for Well №2025 and Well №2026 (Case Study 1).

The current section presents the flow rate and skin factor equations that were implemented in the SkinFlow platform, as they form the core of the computational framework.

The oil flow rate calculated via Darcy's law (Eq. 2) has been fully integrated into the platform:

$$q_0 = \frac{k \cdot k_{ro} \cdot h \cdot (p_r - p_{wf})}{141.2 \cdot \mu_o \cdot B_o \cdot \left( \ln \left( \frac{r_e}{r_w} \right) + S \right)} \quad (2)$$

For horizontal wells, the Darcy framework still applies, but the simple radial flow model used for vertical wells no longer accurately describes the flow geometry. Due to their extensive contact with the reservoir, horizontal wells drain a larger volume than vertical wells, requiring an inflow equation that accounts for

their finite length and permeability anisotropy. In this study, a Joshi-type expression (S.D.Joshi, 1988) is applied for open-hole horizontal wells:

$$q_0 = \frac{k_H \cdot k_{ro} \cdot h \cdot (p_r - p_{wf})}{141.2 \cdot \mu_o \cdot B_o \cdot \left( \ln \left( \frac{a + \sqrt{a^2 - \frac{L^2}{2}}}{L/2} \right) + \frac{\sqrt{\frac{k_H}{k_v}} h}{L} \ln \left( \frac{\sqrt{\frac{k_H}{k_v}} h}{2r_w} \right) + S \right)}, \quad (3)$$

where  $k_H$  and  $k_v$  are horizontal and vertical permeabilities,  $L$  is lateral length,  $a$  is the drainage semi-axis related to the effective radius. The geometric term involving  $a$  captures end effects of finite laterals, while the anisotropy ratio  $\beta = \sqrt{k_H/k_v}$  corrects the vertical convergence term.

While Eq. 1 is applied to an open-hole vertical well, the computation of the skin factor for horizontal wells is an adaptation of the traditional vertical Hawkins model. Furui, Zhu, and Hill (2003) showed that near-wellbore damage in anisotropic media forms elliptical, not cylindrical, pressure isobars. Their extension of Hawkins' formula, which was integrated into the platform, yields a local skin of the form:

$$s(x) = \left( \frac{k}{k_d(x)} - 1 \right) \ln \left[ \frac{1}{\sqrt{I_{ani} + 1}} \left( \frac{r_{dH}(x)}{r_w} + \sqrt{\left( \frac{r_{dH}(x)}{r_w} \right)^2 + I_{ani}^2 - 1} \right) \right] \quad (4)$$

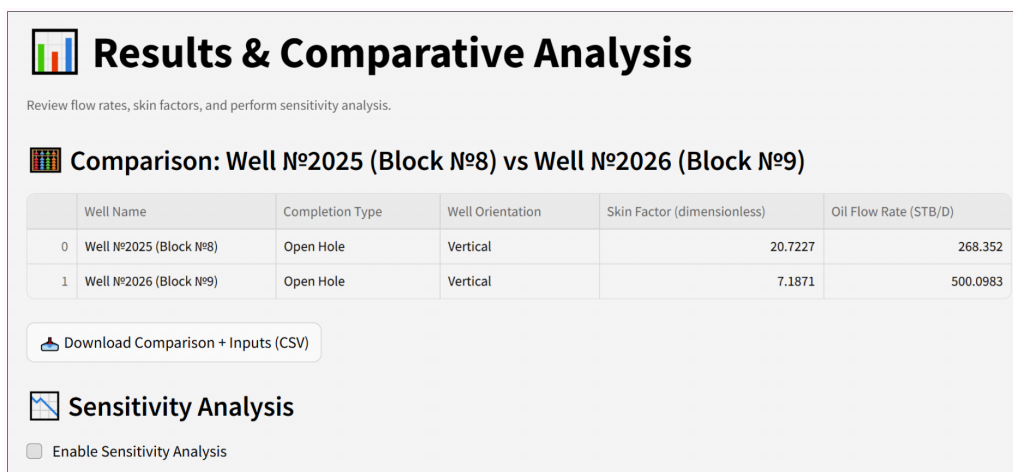
with  $I_{ani} = \sqrt{k_H/k_v}$ . The overall equivalent skin is obtained by integrating these local contributions along the lateral, where  $k_d(x)$  is the damaged-zone permeability distribution and  $r_{dH}(x)$  is the horizontal semi-axis of the damaged ellipse at position  $X$ . Thus, both the productivity equation and the skin model explicitly incorporate permeability anisotropy and finite-length effects, which are essential corrections when modeling open-hole horizontal inflow. These principles are applied in the following case study, which uses a dual-well comparison to provide the quantitative analysis needed for a critical design decision.

**Case Study 1: Dual-Well Comparison for Design Decision.** In this scenario, the engineer is responsible for evaluating two recently drilled vertical wells, both completed in open hole style and intersecting the same formation. The formation and fluid conditions were identical across both wells, including reservoir permeability, thickness, reservoir pressure, and fluid viscosity. However, the wells differ in terms of near-wellbore damage (Fig. 4).

Well №2025 exhibits more severe damage conditions, with an altered zone permeability of only 10 mD and a radius of 8 ft. In contrast, Well №2026 shows improved near-wellbore permeability ( $k_s = 25$  mD), but a slightly larger radius of the altered zone of 12 ft. Both wells share the same wellbore radius and fluid expansion factor.

The goal was to evaluate whether stimulation of Well №2025 could significantly reduce formation damage and improve production performance, bringing it closer to the benchmark set by the more productive Well №2026.

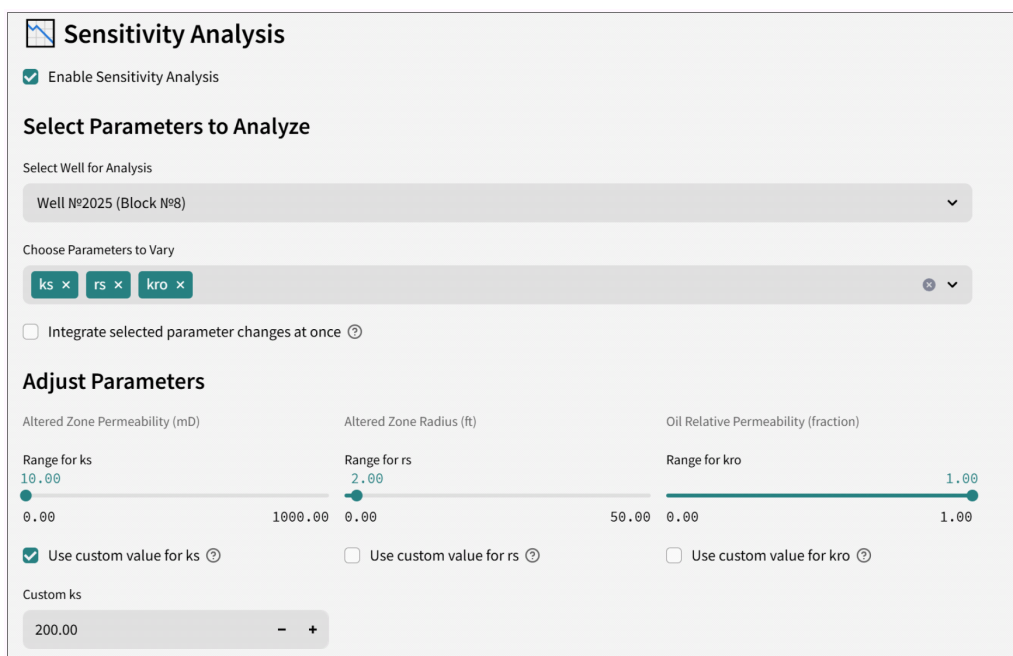
The dual-well comparison module computes the oil flow rate and corresponding skin factor for each well (Fig. 5). Results indicate that Well №2026 initially outperforms Well №2025, producing 500.1 STB/D with a skin factor of 7.19, compared to 268.35 STB/D and a skin factor of 20.72 for Well №2025.



**Figure 5—Results and comparative analysis interface showing computed skin factor and oil flow rate for Well №2025 (Block №8) and Well №2026 (Block №9).**

To explore the potential for performance improvement in Well №2025 through targeted stimulation, a sensitivity analysis was conducted.

**Sensitivity Analysis Results.** To explore parameter sensitivity, the platform gives engineers the flexibility to select input variables from dropdown menus and adjust them using sliders (Fig. 6). Each parameter can be modified independently while the others remain fixed, or multiple adjustments can be integrated at once to evaluate the combined effect on well performance. The sliders have fixed maximum values to maintain consistency, but when engineers need to test conditions outside those limits — or simply prefer not to use the slider — they can switch to the custom value option. Small question mark icons placed next to each input provide additional guidance, helping the user understand how to apply manual entries.



**Figure 6—Parameter selection and slider interface for Well №2025. The user chose to view parameter effects separately and quickly inserted a custom value for altered zone permeability.**

Sensitivity analysis was performed on Well №2025 (Block №8) by enabling three key variables that can be influenced through stimulation:



- Altered Zone Permeability ("ks")
- Altered Zone Radius ("rs")
- Oil Relative Permeability ("kro")

When "ks" was increased from 10 to 200 mD (representing high-efficiency acidizing), flowrate increased from 268.35 STB/D to 1207.04 STB/D — a +349.8% improvement. Reducing "rs" from 8 ft to 2 ft led to a flowrate of 388.07 STB/D (+44.6%), while increasing "kro" from 0.85 to 1.00 brought production up to 315.71 STB/D (+17.6%) (Fig. 7).

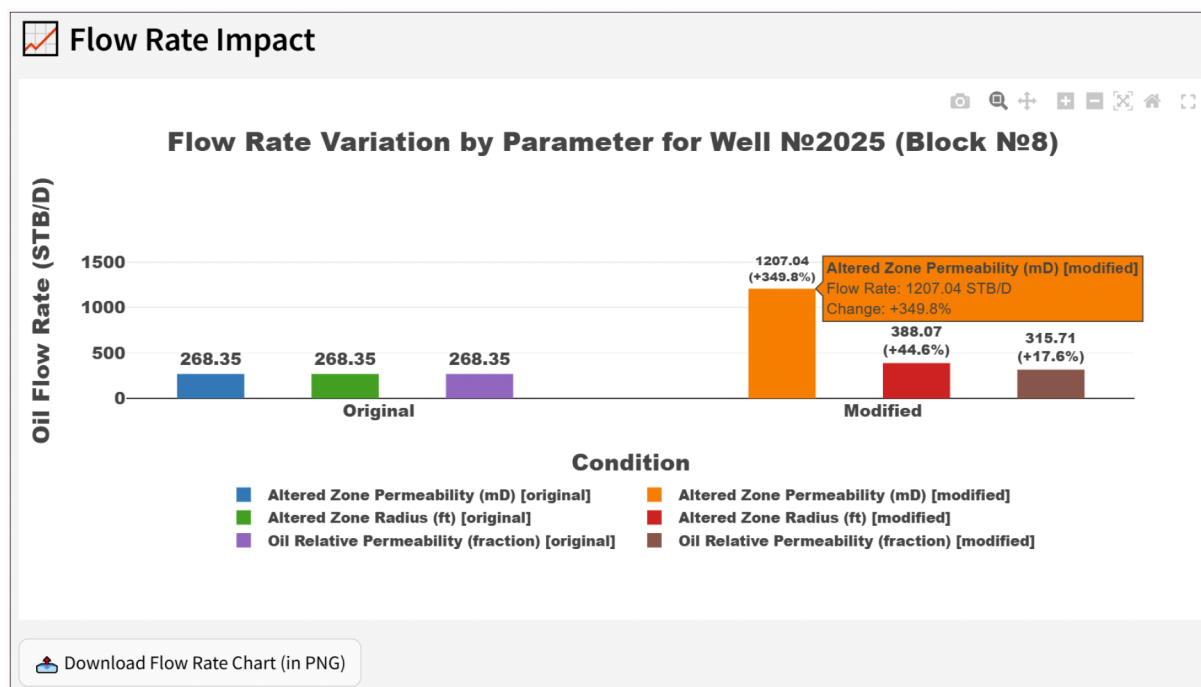


Figure 7—Flow rate impact by varying "ks", "rs", and "kro" for Well №2025. The chart shows original vs. modified flow rates, with percentage changes clearly labeled above each bar. A download button is available below the chart for saving the figure for reports

In terms of formation damage, the skin factor dropped significantly when "ks" was increased — from 20.72 to 11.71 (a -43.5% reduction), confirming the effect of potential stimulation on improving near-wellbore permeability (Fig. 8).

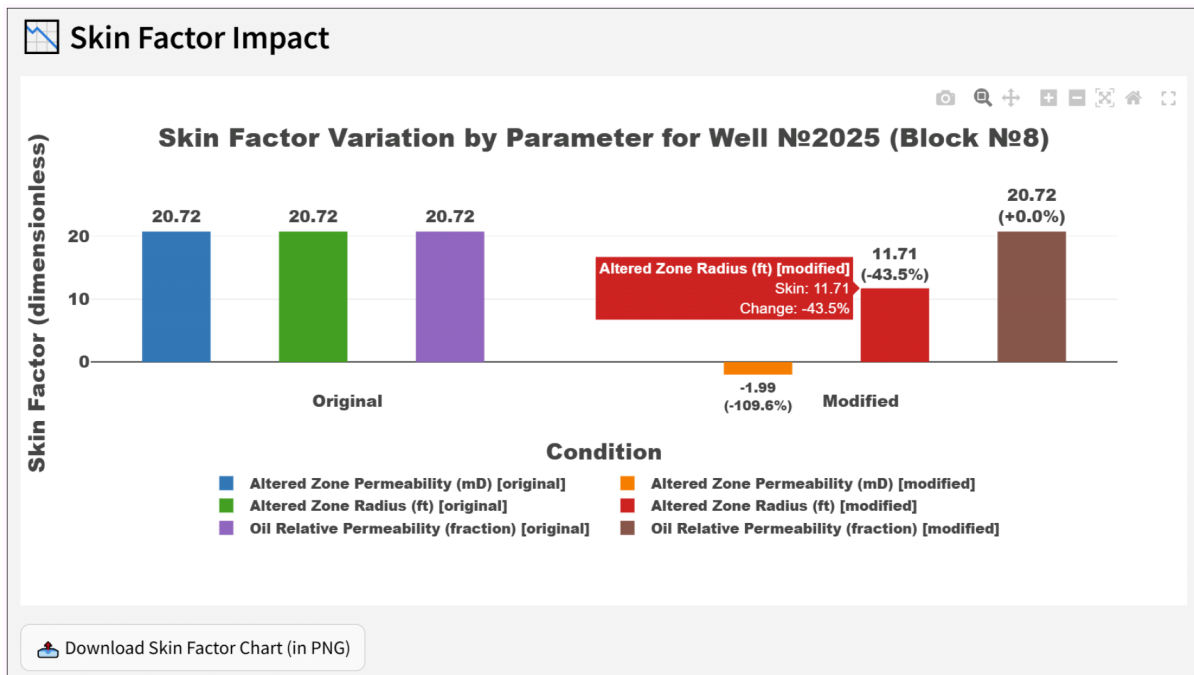


Figure 8—Skin factor impact by varying "ks", "rs", and "kro" for Well №2025. The chart highlights how skin factor decreases with parameter adjustments, with percentage changes labeled above each modified bar. A download button is available for saving the chart.

These changes reflect real-world engineering actions, such as mud cake removal, acid washing, or wettability alteration, suggesting that Well №2025 is a viable candidate for remediation and could achieve substantial performance gains that narrow the gap with Well №2026.

**Monte Carlo Analysis.** To account for operational uncertainty in the post-stimulation state of Well №2025 (Block №8), a Monte Carlo simulation was conducted over 1000 iterations (Fig. 9). The following distributions were applied, reflecting the expected impact of acidizing and near-wellbore cleanup:

- "ks": lognormal (mean = 100, std dev = 25)
- "rs": triangular (min = 1, mode = 3, max = 6)
- "kro": normal (mean = 0.85, std dev = 0.04)

Select Well for Analysis

Well №2025 (Block №8)

### Monte Carlo Simulation

Permeability parameters like  $k$ ,  $k_s$ ,  $k_{RH}$ , etc. default to lognormal distribution.

Define Distributions

Select Parameters for Simulation

Altered Zone Per... x Altered Zone Ra... x Oil Relative Per... x

Distribution Type for Altered Zone Permeability (mD)

lognormal

ks Mean (natural scale)

100.00

ks Std Dev (natural scale)

25.00

Distribution Type for Altered Zone Radius (ft)

triangular

rs Min

1.00

rs Mode

3.00

rs Max

6.00

Distribution Type for Oil Relative Permeability (fraction)

normal

kro Mean

0.85

kro Std Dev

0.04

Number of Simulations

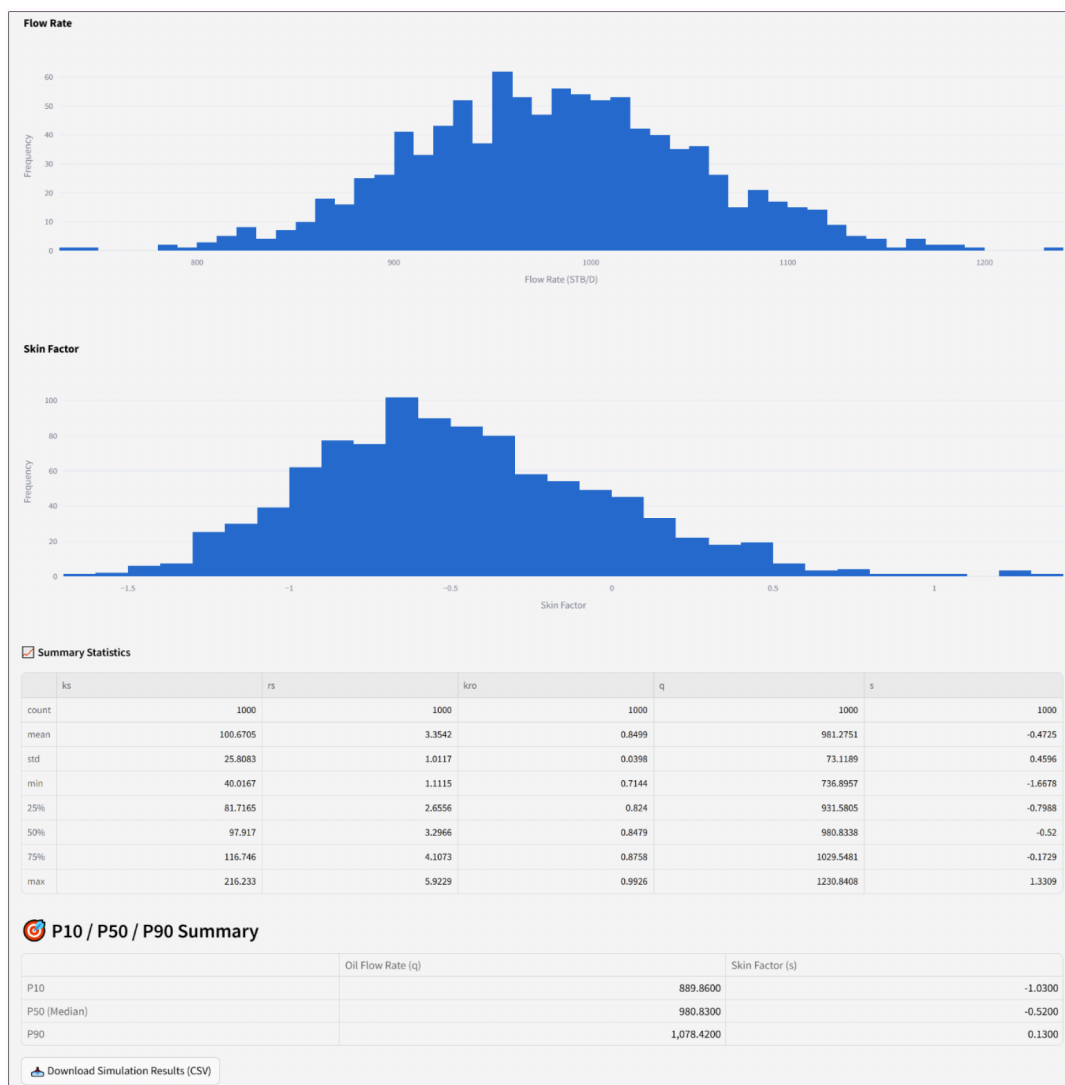
1000

Run Monte Carlo Simulation

**Figure 9—Monte Carlo input distributions for well №2025. The engineer selected  $k_s$ ,  $r_s$ , and  $k_{ro}$  as uncertain variables, each assigned realistic probability distributions before running 1000 simulations.**

The simulation outcomes (Fig. 10) show that post-stimulation flow rates are expected to cluster tightly around high values. Oil flow rate spans from a P10 of 890 STB/D to a P90 of 1078 STB/D, with a median of ~981 STB/D. The corresponding skin factor improves dramatically, ranging between -1.0 and +0.1, with a median of about -0.5, pointing to effective removal of near-wellbore damage.

These results emphasize that stimulation of Well №2025 is not only likely to succeed but is also robust under uncertainty: even under conservative assumptions (P10 case), flow rate exceeds three times the original pre-treatment production of 268 STB/D. From an operational standpoint, this confirms that Well №2025 represents a strong candidate for stimulation, and that the intervention could close much of the productivity gap with Well №2026, while offering a favorable risk-to-reward balance.



**Figure 10—Histogram results and summary statistics from Monte Carlo Simulation. Histograms show the distribution of oil flow rate (q) and skin factor (s) across 1000 runs, with summary statistics and P10–P50–P90 values. The median outcome indicates ~981 STB/D and a skin of –0.52, confirming strong stimulation response.**

Although Well №2026 (500 STB/D at a skin factor of 7.19) initially outperformed Well №2025 by a wide margin, the updated simulation shows that stimulation could transform Well №2025 into a nearly comparable producer (Table 1). With post-stimulation flow rates clustering around 981 STB/D (median) and skin factors close to zero, the well more than triples its original output of 268 STB/D and, in favorable cases, even surpasses the benchmark set by Well №2026. This places Well №2025 into a much more productive range, especially in areas where surface infrastructure is already in place.

**Table 1—Well Performance Parameters Before and After Stimulation of Well №2025, Including Benchmark Data from Well №2026**

| Parameter                      | Before Stimulation | After Stimulation, Deterministic | After Stimulation, Monte Carlo | Well 2026 |
|--------------------------------|--------------------|----------------------------------|--------------------------------|-----------|
| Altered Zone Permeability (mD) | 10                 | 200                              | 100 (mean)                     | 25        |
| Skin Factor                    | 20.72              | 11.71                            | -0.52 (median)                 | 7.19      |
| Oil Flow Rate (STB/D)          | 268.3              | 1207.04                          | 980.8 (P50)                    | 500.1     |

**Case Study 2: Stimulation Planning for Pilot Horizontal Well №872\_H.** In this scenario, the engineering team faced a decision regarding the pilot open-hole horizontal well №872\_H, drilled into a tight, low-permeability formation. The initial well test confirmed limited productivity, with an inflow of 120.6 STB/D and a computed skin factor of 13.95 along with analysis of buildup data indicating a matrix permeability of only 10 mD. Test interpretation further revealed strong anisotropy, with  $k_H$  around 10 mD and  $k_V$  near 1 mD, consistent with log-derived facies correlations. Additional near-wellbore damage reduced the altered-zone permeability to 2 mD. These results raised the question of whether stimulation could turn the pilot well into a viable producer.

For evaluation, the stimulation team entered the pilot well data into the SkinFlow platform, with all input parameters summarized in Fig. 11.

| Section                              | Parameter                                  | Value     |
|--------------------------------------|--------------------------------------------|-----------|
| Well Completion Type                 | Enter Name for Well                        | 872_H     |
|                                      | Select Well Completion Type                | Open Hole |
| Reservoir & Fluid Parameters         | Reservoir Permeability [k] (mD)            | 10.00     |
|                                      | Permeability of the altered zone [ks] (mD) | 2.00      |
|                                      | Wellbore Radius [rw] (ft)                  | 0.30      |
|                                      | Radius of the altered zone [rs] (ft)       | 10.00     |
|                                      | Reservoir Thickness [h] (ft)               | 85.00     |
|                                      | Reservoir Pressure [p_res] (psi)           | 4000.00   |
|                                      | Well Flowing Pressure [p_wf] (psi)         | 2800.00   |
|                                      | Oil Viscosity [μo] (cP)                    | 2.50      |
| Well Orientation Specific Parameters | Oil Relative Permeability (fraction)       | 0.80      |
|                                      | Formation Volume Factor [Bo] (res bbl/STB) | 1.20      |
|                                      | Drainage Radius [re] (ft)                  | 1500.00   |
|                                      | Horizontal Section Length [L] (ft)         | 2000.00   |
|                                      | Horizontal Permeability [kH] (mD)          | 10.00     |
|                                      | Vertical Permeability [kV] (mD)            | 1.00      |

Figure 11—Input parameters for pilot horizontal well №872\_H. All reservoir, fluid, and horizontal-well specific data are entered in structured fields, including horizontal section length, and anisotropic permeabilities ( $k_H$ ,  $k_V$ ).

**Sensitivity Analysis with Integrated Changes.** To evaluate the stimulation potential, the engineer activated the "Integrate selected parameter changes at once" feature in SkinFlow, combining the effect of two operational changes (Fig 12):

- Near-wellbore stimulation represented by an increase of altered-zone permeability ( $k_s$ ) from 2 mD to 10 mD.
- Drawdown enhancement by lowering flowing pressure ( $p_{wf}$ ) from 2800 psi to 2400 psi, entered via the custom value option on the slider to mimic the capacity of a stronger downhole pump.



ks x p\_wf x

☒ Integrate selected parameter changes at once ?

### Adjust Parameters

Altered Zone Permeability (mD)

Well Flowing Pressure (psi)

Range for ks

Range for p\_wf

10.00

2800.00

0.001000.00

0.0010000.00

☐ Use custom value for ks ?

☒ Use custom value for p\_wf ?

Custom p\_wf

2400.00 - +

Figure 12—The engineer selected both altered-zone permeability and flowing pressure to be varied together. After setting a custom value of 2400 psi for p\_wf and ks via the slider, the "Integrate selected parameter changes at once" option was enabled to observe their combined impact.

The resulting bar charts (Figs. 13–14) illustrate the combined effect of these modifications. Flow rate increased more than tenfold, from 120.6 STB/D to 1274.5 STB/D (+956.6%), while the skin factor dropped from 13.95 to nearly zero (–100%).

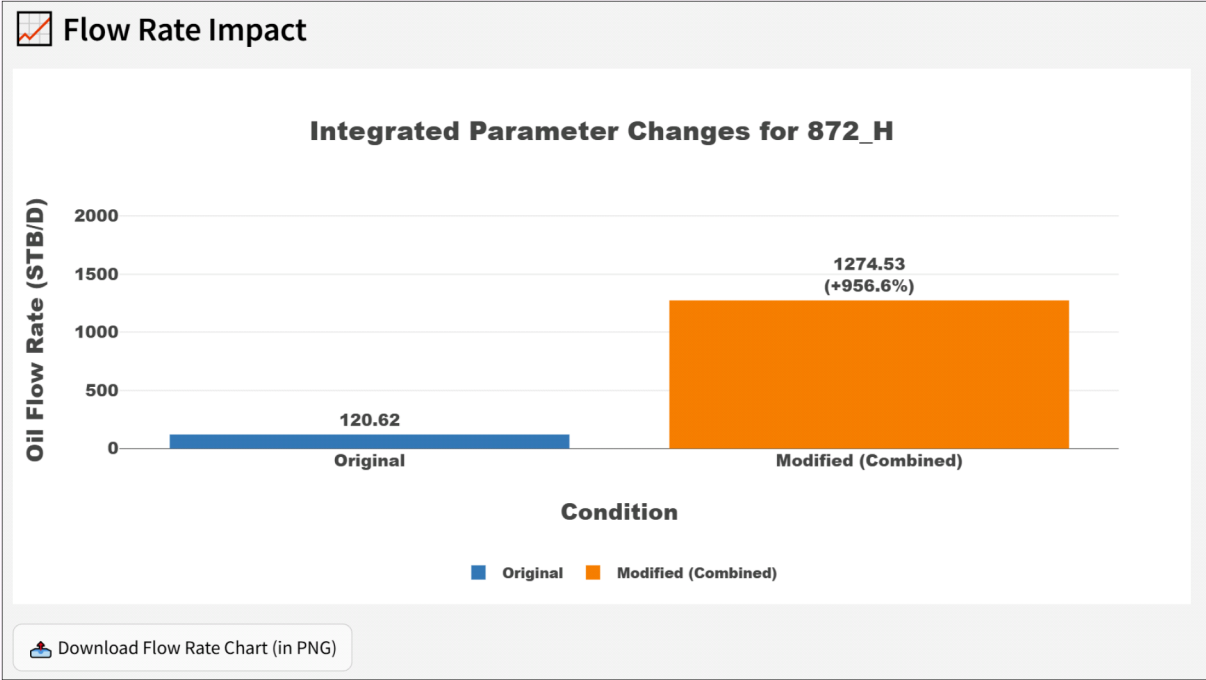


Figure 13—Flow rate impact with integrated parameter changes for Well №872\_H. The oil rate increased nearly tenfold, from 120.62 to 1274.53 STB/D, after stimulation.

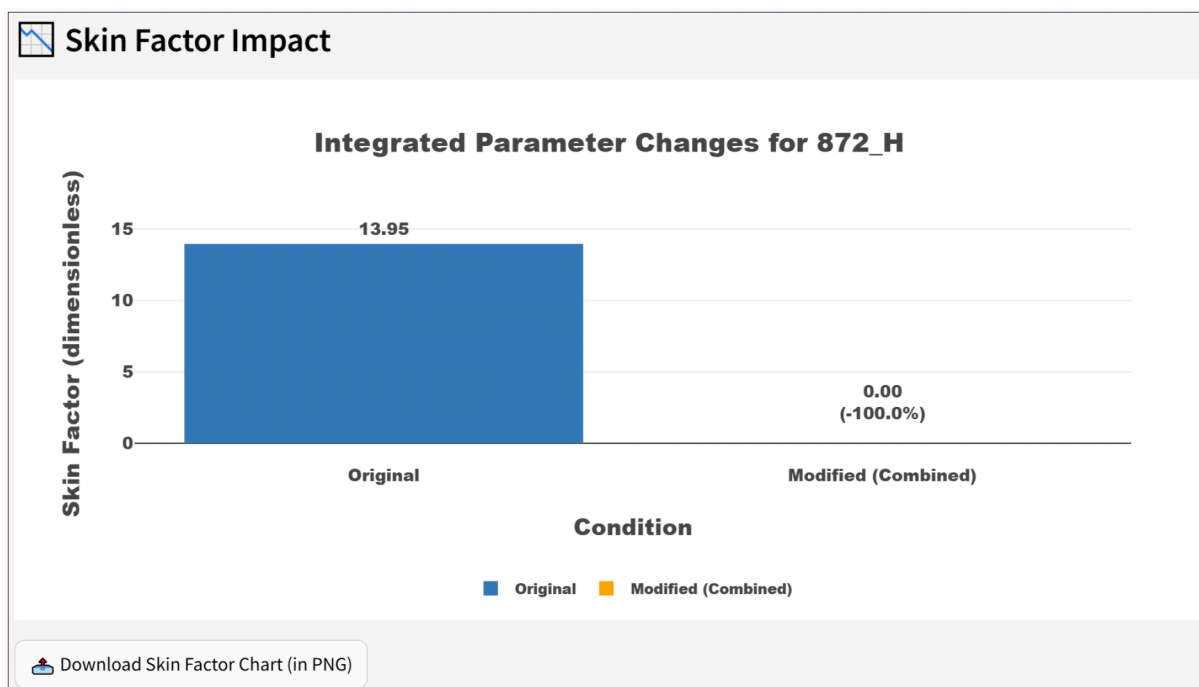


Figure 14—Skin factor impact with integrated parameter changes for Well №872\_H. The skin decreased from 13.95 to 0.00 (–100%), confirming that stimulation fully removed near-wellbore damage.

The results clearly demonstrated that stimulation of Well №872\_H would unlock significant productivity gains, transforming the pilot from a marginal to a commercial well. By combining stimulation design (ks improvement) with pump capacity planning (BHP reduction), the engineering team could anticipate over 10× higher production rates alongside the elimination of formation damage effects. A more robust Monte Carlo analysis, which integrated uncertainty in both parameters, yielded a most-likely (P50) outcome of 839 STB/D at a skin factor of 1.02, reflecting a more conservative but realistic forecast of the post-stimulation performance.

Table 2—Before and After Stimulation Results for Well №872\_H

| Parameter                                | Before Stimulation | After Stimulation, Deterministic | After Stimulation, Monte Carlo |
|------------------------------------------|--------------------|----------------------------------|--------------------------------|
| Altered Zone Permeability (ks, mD)       | 2                  | 10                               | 8 (mean)                       |
| Flowing Pressure (p <sub>wf</sub> , psi) | 2800               | 2400                             | 2400 (mean)                    |
| Skin Factor (dimensionless)              | 13.95              | 0                                | 1.02 (median)                  |
| Oil Flow Rate (STB/D)                    | 120.62             | 1274.53                          | 839.36 (P50)                   |

**Understanding Model Parameters in Terms of Field Conditions.** In applying SkinFlow, it is important to note that the parameters adjusted in the system are not abstract values but direct representations of measurable field conditions. Furthermore, the mathematical models that process these parameters are all drawn from peer-reviewed academic literature. Near-wellbore damage caused by drilling mud invasion or fines migration is captured by a reduction in altered-zone permeability ( $k_s$ ) relative to the undisturbed reservoir permeability ( $k$ ), as well as by the radial extent of the damaged region ( $r_s$ ). Stimulation treatments such as acidizing translate into higher values of  $k_s$  and, in some cases, smaller values of  $r_s$ , reflecting a cleaner and more conductive near-wellbore environment. Relative permeability to oil ( $k_{ro}$ ) captures fluid mobility and saturation effects, making it responsive to conditions such as water cut or wettability alteration. An increase in  $k_{ro}$  therefore represents improved oil mobility following treatment. Beyond the

near-wellbore region, reservoir and operational parameters can also be modified to represent field actions. For example, reservoir pressure ( $p_{res}$ ) may decline with depletion or be sustained by water or gas injection, while bottomhole flowing pressure ( $p_{wf}$ ) can be lowered by installing artificial lift or optimizing choke settings. Drainage radius ( $r_e$ ) reflects the spacing of producing wells and can change with infill drilling or boundary redefinition. Reservoir thickness ( $h$ ) may be adjusted to represent effective pay after petrophysical re-evaluation. On the fluid side, oil viscosity ( $\mu_o$ ) and formation volume factor ( $Bo$ ) may be updated with new PVT analysis or after treatment that alters fluid properties.

For horizontal wells, two distinct corrections are applied. The inflow equation follows Joshi (1988), which replaces the vertical  $\log(r_e/r_w)$  term with a denominator that accounts for finite lateral length and permeability anisotropy. In parallel, the skin factor is evaluated using the Furui–Zhu–Hill approach (2003), which generalizes Hawkins' vertical model to anisotropic horizontal completions by treating near-wellbore damage as elliptical rather than cylindrical. In this way, the platform translates a wide range of real-world conditions — mud damage, reservoir stimulation, pressure management, artificial lift, or well spacing — into intuitive parameters that can be directly supplied as inputs. This linkage ensures that the tool remains relevant both as an engineering decision aid and as a teaching platform with direct practical applicability.

## Conclusions

SkinFlow fills a gap in well performance analysis with an interactive, model-based tool that shows how near-wellbore conditions affect skin factor and oil flow through single- and dual-well evaluations, parameter-based forecasts, and instant visual feedback.

By varying parameters individually or in combination, the system illustrates how drilling damage or stimulation treatments affect production. The Monte Carlo module extends this by quantifying uncertainty, giving engineers realistic outcome ranges to guide risk-aware decision making. In practice, these features support planning of remedial treatments, optimization of drawdown strategies, and prioritization of wells for stimulation.

Beyond field application, SkinFlow also serves an educational role, linking theory with real-time interaction. Students and trainees can see how reservoir, fluid, and well-orientation parameters influence performance, building a deeper understanding of well hydraulics and damage modeling.

Looking forward, the platform can be extended with features to strengthen its role as a decision-support system. Potential additions include support for another completion types (cased and perforated, slotted liners, gravel packs, or complex multilateral geometries), modules for economic feasibility analysis, productivity index tracking, and AI-based recommendations informed by historical well data. An important step is well integrity evaluation before stimulation, as defects such as flow behind casing, poor isolation, or mechanical damage can limit a well's suitability for treatment, and integrity checks give engineers greater confidence in decisions. Another area of development is the integration of physics-informed equations, similar to those used in commercial reservoir simulators, to expand predictive capability while retaining its interactive and lightweight character.

In conclusion, SkinFlow provides a flexible and technically sound framework that links measurable reservoir and well parameters to production outcomes. It offers petroleum engineers and students alike a practical environment to explore, understand, and optimize well performance in both vertical and horizontal open-hole completions.

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